Murad Hossain

CFD, Fluid Mechanics & Computational Mechanics | Research-focused academic profile

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Research direction

My academic work examines a modified inverted BE50 airfoil and lightweight structural design. I am interested in credible numerical methods, mesh convergence, validation and the connection between physics and geometry.

Education and English qualification

BSc in Mechanical Engineering — Rajshahi University of Engineering & Technology (RUET), 2019. IELTS Academic overall 7.5 (Listening 8.5, Reading 7.5, Writing 6.5, Speaking 6.5).

Track-relevant engineering evidence

• Modified flipped BE50 airfoil: lift and drag simulation: Undergraduate researcher. RUET undergraduate thesis, 2019; subsequent draft needs independent validation and grid-sensitivity work.

• Trailing-edge inward dimples and passive flow control: Simulation and research drafting. Draft stage; mesh independence and external validation remain incomplete.

• FEA-driven topology optimization and geometry reconstruction: Co-author; precise contributions to confirm. 446 g → 242 g (45.7% modeled mass reduction); minimum static safety factor approximately 1.974.

Research records

• Simulation of Modified Flipped Airfoil and Analysis of Lift and Drag Coefficient — BSc thesis · RUET · 2019.

• Effect of trailing-edge inward dimples on the aerodynamic performance of an inverted low-Reynolds-number BE50 airfoil — Draft manuscript; not published.

• FEA-Driven Topology Optimization for Lightweight Structural Design: A Computational Framework with Implicit Geometry Reconstruction — Accepted conference paper · ICMIME 2026 · ID 41.

Relevant credentials

SOLIDWORKS CAD Design Associate · SOLIDWORKS Additive Manufacturing Associate · 3DSwymer Associate · SOLIDWORKS Sustainability Associate

Technical skills

ANSYS Fluent · CFD meshing · Lift/drag analysis · FEA · SolidWorks · Topology optimization

Employment context

Walton Hi-Tech Industries PLC (May 2021–present); New Model Boiler Company Limited (Nov 2019–Apr 2021). Full employment records and references available on request.